MAHATMAGANDHIUNIVERSITY,KOTTAYAM
MGU-UGP(HONOURS) REGULAR EXAMINATION NOVEMBER 2025

THIRD SEMESTER
Discipline Specific Elective (DSE)-MG3DSEBBT201 - Developmental Biology and Assisted Reproduction Technology
(2024ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A- One word answer-Answer any10 questions. Each question carries 2 marks

1. What are the two primary functions of male gonad? [K]/CO1
2. Name any two hormones produced by ovary. [K]/CO1
3. List out any two examples of types of egg based on the amount of yolk. [K]/CO1
4. Which muscles contract during childbirth? [U]/CO2
5. Which placental hormone maintains the endometrium? [U]/CO2
6. What are the first two stages of parturition? [U]/CO2
7. List out any two morphogenetic movements [U]/CO3
8. Write two Examples of organisms withmeroblastic cleavage? [U]/CO3
9. Which type of cleavage occurs in humans? [U]/CO3
10. Write the expansion ofIVF. [An]/CO4
11. Write any two techniques used for Prenatal Diagnosis [U]/CO4
12. What is the 16-cell stage called in embryo development? [K]/CO2

(10 x 2 = 20)

Part B-Short essay-Answer any 4 questions. Each question carries 5 marks

13. Name the different types of eggs based on yolk and define each briefly [K]/CO1
14. Describe a diagram showing the process of oogenesis with its stages [K]/CO1
15. What is ectopic pregnancy? Name two sites where it may occur [K]/CO2
16. Describe the term polyspermy and how it is prevented in humans [U]/CO2
17. Classify morphogenetic movements with suitable examples [U]/CO3
18. Discuss the emotional and ethical concerns involved in prenatal diagnosis [An]/CO4

(4 x 5= 20)

Part C-Essay-Answer any 2 questions. Each question carries 15 marks

19. Analyze the role of FSH, LH, estrogen, and progesterone in regulating the female reproductive cycle [K]/CO1
20. Describe the development and functions of the placenta in supporting pregnancy [U]/CO2
21. Show the sequential developmental stages from cleavage to germ layer formation using labeled diagrams and a flowchart [U]/CO3
22. Explain a detailed diagrammatic representation showing the steps of IVF and embryo transfer [U/CO4]

(2X15=30)